JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester: CIE-3/Test-2

Discrete Mathematical Structures

Duration : 1Hr.15Mins Date: 09-01-2026 Max. Marks: 30

Subject Code:24CS350

NOTE: Part-A is compulsory, and Part-B has choice.

Q.NO	Scheme of Evaluation	Marks	Total Marks
	PART-A		
1.	$A=\{1,2,3\}$ $B=\{w,x,y,z\}$ $C=\{4,5,6\}$ $R_1=\{(1,w)(3,w)(2,x)(1,y)\}$ $R_2=\{(w,5)(x,6)(y,4)(y,6)\}$ $R_3=\{(w,4)(w,5)(y,5)\}$		08
	a) $R_1\circ(R_2\cup R_3)=(R_1\circ R_2)\cup(R_1\circ R_3)$	02	
	$(R_2\cup R_3)=\{(w,4)(w,5)(x,6)(y,4)(y,5)(y,6)\}$ $R_1\circ(R_2\cup R_3)=\{(1,4)(1,5)(3,4)(3,5)(2,6)(1,6)\}$ $(R_1\circ R_2)=\{(1,5)(3,5)(2,6)(1,4)(1,6)\}$ $(R_1\circ R_3)=\{(1,4)(1,5)(3,4)(3,5)\}$ $(R_1\circ R_2)\cup(R_1\circ R_3)=\{(1,4)(1,5)(1,6)(2,6)(3,4)(3,5)\}$. Therefore $R_1\circ(R_2\cup R_3)=(R_1\circ R_2)\cup(R_1\circ R_3)$ is TRUE		
	b) $R_1\circ(R_2\cap R_3)=(R_1\circ R_2)\cap(R_1\circ R_3)$.	02	
	$(R_2\cap R_3)=\{(w,5)\}$. $R_1\circ(R_2\cap R_3)=\{(1,5)(3,5)\}$ $R_1\circ R_2=\{(1,5)(3,5)(2,6)(1,4)(1,6)\}$ $R_1\circ R_3=\{(1,4)(1,5)(3,4)(3,5)\}$. $(R_1\circ R_2)\cap(R_1\circ R_3)=\{(1,4)(1,5)(3,5)\}$. Therefore $R_1\circ(R_2\cap R_3)=(R_1\circ R_2)\cap(R_1\circ R_3)$ is FALSE		

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c) $M(R_1 \circ R_2) = M(R_1) \circ M(R_2)$ — 02

$$M(R_1 \circ R_2) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{ccc} 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$M(R_1) \cdot M(R_2) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} w & x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \end{array} \cdot \begin{array}{c} \\ w \\ x \\ y \\ z \end{array} \begin{array}{c} \begin{array}{ccc} 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \end{array} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Therefore $M(R_1 \circ R_2) = M(R_1) \circ M(R_2)$ Is TRUE

d) $M(R_1) \circ M(R_2) \; ! = M(R_2) \circ M(R_1)$ — 02

$$M(R_1) \cdot M(R_2) = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$M(R_2) \cdot M(R_1) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore M(R_1) \circ M(R_2) \; ! = M(R_2) \cdot M(R_1)$ Is TRUE.

2 — Explanation on each type of Graph and suitable example. — 07

a) Complement Graph – 1½
b) Subgraph – 1½
c) Euler Graph – 1½
d) Hamiltonian-Graph – 1½
e) Trail and path – 1.

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Q.NO	Scheme of Evaluation	Marks	Total Marks
	PART-B		
3	$A = \{1,2,3,4,5,6,7,8,9,10,11,12\}$. $(x,y) \in R$, iff $(x-y)$ is Multiple of 5		
a)	Verify whether R is equivalence relation	3½	

$R = \{(1,1)(1,6)(1,11)$
$(2,2)(2,7)(2,12)$
$(3,3)(3,8)$
$(4,4)(4,9)$
$(5,5)(5,10)$
$(6,6)(6,1)(6,11)$
$(7,7)(7,2)(7,12)$
$(8,8)(8,3)$
$(9,9)(9,4)$
$(10,10)(10,5)$
$(11,11)(11,1)(11,6)$
$(12,12)(12,2)(12,7)\}$.

i) $\forall x \in A\ (x,x) \in R$ therefore Reflexive.

i.e $x-x = 0 = 5k$ where $k=0$.

ii) $\forall (x,y) \in R$, $(x,y) \in R$ then $(y,x) \in R$.

therefore ~~reflex~~ Symmetric.

if $(x-y) = 5k$ for some integer k.

then $(y-x) = -5k$.

which is also Multiple of 5.

∴ R is symmetric.

Eg: $(1,6)(6,1)(2,7)(7,2)(2,12)(12,2)$

iii) $\forall (x,y) \in R$, if $(x,y) \in R$ and $(y,z) \in R$.

then $(x,z) \in R$.

Eg: $(2,12)(12,7)$ then $(2,7)$.

$2-12 = 10/5 = 2$, $12-7 = 5/5 = 1$, $2-7 = 5/5 = 1$

∴ Transitive.

∴ As R is Reflexive, Symmetric & Transitive, R is an Equivalence Relation on A

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b) If yes, Find the partition of A induced by R. 3½

Find Equivalence classes.

$[1] = \{1, 6, 11\}$ $[6] = \{6, 1, 11\}$

$[2] = \{2, 7, 12\}$ $[7] = \{7, 2, 12\}$

$[3] = \{3, 8\}$ $[8] = \{8, 3\}$

$[4] = \{4, 9\}$ $[9] = \{9, 4\}$

$[5] = \{5, 10\}$ $[10] = \{10, 5\}$.

∴ The partitions are.

$P = \{[1], [2], [3], [4], [5]\}$

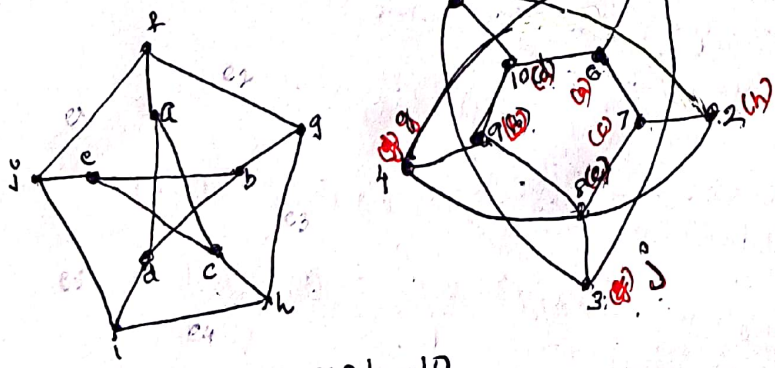
because $[1] \cup [2] \cup [3] \cup [4] \cup [5] = A$

and $[1] \cap [2] \cap [3] \cap [4] \cap [5] = \phi$

4 Explanation of Isomorphism in graphs. 02

Verifying Isomorphism. 03

Mapping of Vertices and Edges 03



$|V(a)| = 10$ $|V(b)| = 10$

$|E(a)| = 15$ $|E(b)| = 15$

degree sequence of a = 3, 3, 3, 3, 3, 3, 3, 3 ... 3.

degree sequence of b = 3, 3, 3, 3 ..... 3.

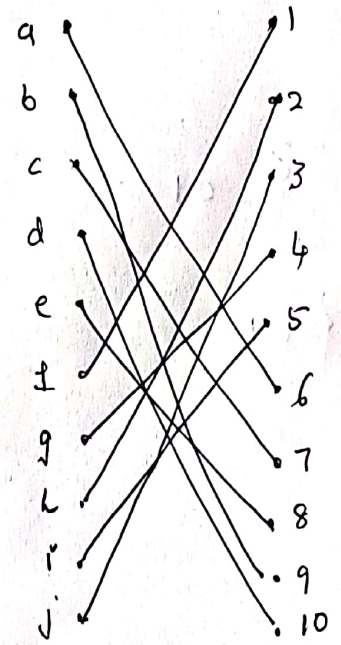
Cycles in graph a: 6 cycles of length 5

Cycles in Graph b: 6 cycles of length 5

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Vertex Mapping

(a) (b)



Therefore

Edge Mapping.

from a to b

(j, f) - (3, 1)
(f, g) - (1, 4)
(g, h) - (4, 2)
(i, h) = (5, 2)
(j, i) = (3, 5)
(j, e) = (3, 8)
(f, a) = (1, 6)
(b, g) = (9, 4)
(c, h) - (7, 2)
(d, i) - (10, 5)
(d, a) - (10, 6)
(a, e) - (6, 7)
(c, e) - 7, 8)
(e, b) - (8, 9)
(b, d) (9, 10)

There is one to one Mapping b/w Graph a and b.

Therefore they are Isomorphic.

OR.

5 Draw the Hasse diagram for divisibility relation on A.

A = {2, 3, 5, 6, 7, 11; 12, 35, 385}

-03-

R = {(2,2)(2, 6) (2, 12)
(3,3) (3,6)(3, 12)
(5,5) (5, 35)(5, 385)
(6,6)(6, 12)
(7,7) (7, 35) (7, 385)
(11,11)(11, 385)
(12,12)(35, 35)(35, 385)
(385)(385)}

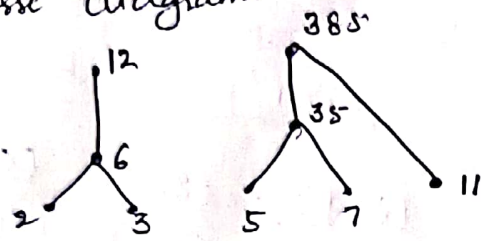
Remove reflexive and transitive relation

After removing the reflexive and transitive relation.

R = {(2, 6)(3, 6)(6, 35) (6, 12)(11, 385)
(7, 35) (35, 385).}

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Hasse diagram.



i) Maximal element : 12, 385
Minimal element : 2, 3, 5, 7, 11 } -01-

ii) Greatest element : Nil
least element : Nil } -01-

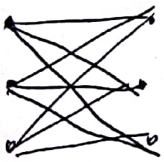
iii) LUB {7, 11} = 385
LUB {5, 7} = 35 } -01-

iv) GLB {2, 12} = 2
GLB {2, 3} = Nil } -01-

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6) Definition of Bipartite Graph with example — 1½
—"— Complete Bipartite Graph —"— 1½

Kuratowski's Graph-1 explanation with Graph $K_{3,3}$ — 2½



$K_{3,3}$.

Kuratowski's Graph-2 explanation with Graph K_5 — 2½

